



Epidemiology of early-onset Parkinson's disease in Finland

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ABSTRACT

Objective: The contribution of genetic causes to Parkinson's disease (PD) is strongest in early-onset cases. We ascertained a nationwide cohort of patients in order to study the genetic epidemiology of early-onset PD (EOPD) in Finland.

Methods: By means of a search in a national database we ascertained all patients with EOPD. These patients had become eligible for reimbursement of PD drugs between the years 1995–2006 and were <55 years of age at the time of PD diagnosis. A total of 441 patients consented and provided clinical and genealogical information.

Results: The incidence of EOPD increased 1.7-fold between the years 1995–2006, the mean annual incidence being 3.3/100,000. Fifty-two patients (11.8%) reported an affected first-degree relative. The birthplaces of patients with PD among first-degree relatives were clustered in certain regions in the southwestern and western coastal provinces of Finland and in the eastern province of Savo. Furthermore, the distance between the birthplaces of the patients' parents was smaller for patients, who had first-degree relatives with PD than for patients with no family history of PD.

Conclusions: Our data suggest that the incidence of EOPD is increasing. The birthplaces of patients with PD among first-degree relatives were clustered in certain provinces of Finland suggesting that monogenic forms of PD or genetic susceptibility of PD are present in the population.

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1. Introduction

The prevalence of Parkinson's disease (PD) is estimated at 300/100,000 (range 31–970/100,000 in registry based or case-finding studies on PD prevalence) in the population [1,2], and, in Finland, a previous epidemiological study has suggested a prevalence of 166/100,000 [3]. The frequency of PD increases with age, as about 1% of individuals over 60 years of age are affected and up to 4% of those over 80 years have PD [1]. Most cases are sporadic, but some 10% of the affected people report a family history of PD [4–8]. Family members of affected patients have 3- to 4- fold higher risk to develop PD than the general population [7,9–11]. Furthermore,

relatives of probands with early-onset PD (EOPD) have higher risk of developing PD than relatives of probands with late-onset PD [12]. Mutations in genes, such as *SNCA*, *PRKN*, *DJ-1*, *PINK1* and *LRRK2* [13], have been identified as a cause of PD, while variation in *SNCA*, *MAPT*, *GBA* and *SMPD1* appears to increase the risk of PD [14].

Small founder population and isolation because of geographic reasons have contributed to the enrichment of certain Mendelian diseases and relative lack of some others [15,16], a phenomenon known as Finnish disease heritage (FDH) [16]. The early settlement in Finland covered the southern and western coastal regions, while late settlement populated the rest of the country. Regional founder effects and isolated small breeding units have shaped the occurrence of diseases belonging to the FDH so that most patients stem from the area of late settlement [16–18].

In the present study we ascertained a nationwide cohort of EOPD patients and studied inheritance modes and geographical distribution of EOPD in Finland.

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2. Subjects and methods

2.1. Study subjects

All permanent residents in Finland are entitled to partial or full reimbursement for prescription medication in many chronic diseases. The reimbursement register is administered by the Social Insurance Institution of Finland (SIIF), which is funded through tax revenues. Medications are reimbursed in full in the case of PD. Reimbursement for PD requires a written statement by a board-certified neurologist that the diagnosis fulfills international criteria for PD.

We ascertained all PD patients, who had become eligible for reimbursement between 1 January 1995 and 31 December 2006, who were <55 years of age at the time of PD diagnosis, and who were alive on 1 October 2007. The diagnoses in ICD-9 and ICD-10 required for reimbursement were searched in the SIIF register leading to the identification of 1090 patients with PD nationwide. Ten children (age 6–15 years), one patient who had died before mailing of the study material and two patients living abroad were excluded. We defined this study population as early-onset PD patients and the incidence rate was calculated based on these 1077 patients.

The study material was mailed to 1077 PD patients. Ascertained patients were invited to fill out a questionnaire and to submit blood for subsequent genetic analysis. Information on current age, age at onset, disease duration and age at L-dopa initiation was requested by means of a structured questionnaire. Family history with respect to PD and dementia was included in the questionnaire in order to determine the frequency of familial cases. In addition, we enquired the birthplaces of the study subjects and their parents.

We received 460 responses giving a response rate of 42.7%. Each participant provided a blood sample and a completed questionnaire. The SIIF then performed another search on the 460 patients in order to confirm the diagnoses of PD. Seven patients were excluded, because other diagnoses had been entertained since the reimbursement decision (isolated dystonia, 2; other degenerative diseases of basal ganglia, 5). Furthermore, 12 patients were excluded on the basis of their responses. Seven patients were excluded, because their responses suggested that their PD diagnosis had been set at age 55 years and five patients were excluded, because other diagnoses had been entertained since the reimbursement decision (progressive supranuclear palsy, 2; drug-induced parkinsonism, 2; undetermined movement disorder, 1). Altogether, 441 patients fulfilled the inclusion criteria of the study.

2.2. Methods

The geographical coordinates of the birthplaces of patients and their parents were determined. ArcGIS software suite (version 10, ESRI, Redlands, CA, U.S.A.) was used to visualize coordinates. We used ESRI (Environmental Systems Research Institute) provided world map templates using WGS84 coordinate system. The geographical coordinates of the birthplaces were placed on the map based on year 2010 municipality information. GeoLocator tool from ArcGIS was used to derive the geographical coordinates in GPS (Global Positioning System) format from birthplace address data. The coordinates that pointed to regions outside current borders of Finland; e.g. the province of Karelia now belonging to Russia, were manually checked from the web service of National Land Survey of Finland (<http://www.maanmittauslaitos.fi/en>). Graphics were drawn using ArcMap (ArcGIS) and kernel density tool to visualize the geographical distribution and density of the birthplaces of the patients. Search radius was set to 2 map units (the shortest width of input feature extents in the output spatial reference) for plotting

the raster image. The analysis was done separately for patients with PD among their first-degree relatives, patients with PD only among their more distant relatives, and patients without PD cases among their relatives.

The distance between the birthplaces of the parents of the patients was calculated, in kilometres, according to an ellipsoid method [19] by using the WGS84 ellipsoid as implemented in the R (R Development Core Team 2011) package *geosphere* (version 1.2–26). Patients with parents birthplace distance over 1160 km were excluded from further analyses, because in such cases at least one of the parents originated from far outside of the study population. The distributions of distances were compared between patients with a positive family history and those with a negative family history. The null hypothesis of equal distributions was tested, by using the two-sided two-sample Kolmogorov–Smirnov (K–S) test. The hypothesis that the parents of patients with positive family history had been born closer to each other than the parents of patients with negative family history was tested by using the one-sided Wilcoxon rank-sum test with continuity correction. The distance distributions were compared graphically by plotting their quintiles against each other as Q–Q plots.

Data obtained from the SIIF were used to calculate annual incidences of EOPD. Incidence was calculated based on the 1090 EOPD patients identified through the national registry. Patients under the age of 16 years ($N = 10$) were excluded as explained above. In addition, annual incidence of PD was calculated on the basis of data obtained from Kelasto, a statistical database maintained by the SIIF (<http://www.kela.fi/in/internet/english.nsf/alias/kelasto>). This database contains all newly diagnosed Finnish PD patients for each year based on the reimbursement decisions. Until the year 2005 the diagnoses of PD and those of PD related disorders (G23, G24 and G90.3) have been lumped in the database and only thereafter the diagnoses have been listed separately. Each year between the years 2005–2011, the PD-related disorders consisted less than 0.1% of the total number of PD and related movement disorders reimbursement decisions.

2.3. Standard protocol approvals, registrations, and patient consents

Permission for the register retrieval was obtained from the SIIF. The study has been approved by the Ethics Committee of Turku

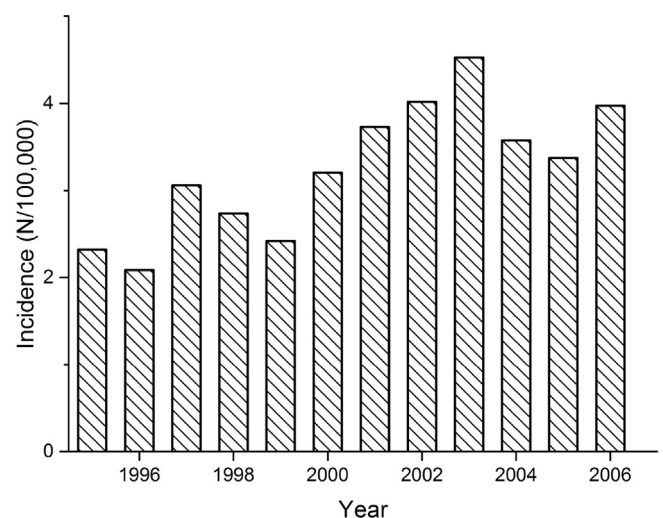


Fig. 1. Incidence of early-onset Parkinson disease in Finland between the years 1995–2006.

Table 1
Clinical features of EOPD patients.

	Total	AAO <50 y	AAO 50–54 y	Familial PD ^a	Sporadic PD
Patients, N	441	273	162	52	306
Men, % (N)	54.9 (242)*	57.1 (156)*	51.9 (84)	55.8 (29)	54.9 (168)
Current age, years	55.1 (5.7)	53.0 (5.7)	58.6 (3.4)	56.5 (5.3)	55.0 (5.9)
Age at symptom onset, years	46.8 (5.5)	44.0 (5.0)	51.5 (1.2)	47.1 (5.1)	46.6 (5.7)
Age at diagnosis, years	48.7 (4.9)	46.4 (4.8)	52.5 (1.1)	49.0 (4.5)	48.5 (5.2)
Age at medication initiation, years	49.0 (4.8)	46.7 (4.9)	52.6 (1.1)	49.3 (4.3)	48.7 (5.2)
Age at L-dopa initiation, years	50.3 (5.0)	48.3 (5.2)	53.5 (2.0)	50.3 (4.6)	50.1 (5.4)
Disease duration, years	8.4 (4.0)	9.0 (4.2)**	7.1 (3.4)**	9.4 (4.8)	8.3 (3.9)
Current L-dopa users, % (N)	76.0 (329)	74.7 (201)	78.1 (125)	84.8 (44)	76.3 (228)

^a Familial PD includes probands who had at least one first-degree relative affected by PD. Probands who had second or third-degree relative affected by PD are not included in this table. Although exact AAO was not available for eight patients, two of them could be confidently classified to the age groups defined in the table. Values are means (standard deviations) if not stated otherwise. * $p < 0.05$, ** $p < 0.001$. AAO, age at onset.

University Hospital. Participating subjects provided written informed consent.

2.4. Statistics

An unadjusted segregation ratio was calculated as the ratio of affected persons to all persons in each group of relatives, excluding the probands. SPSS 17.0 statistical software (IBM SPSS, Chicago, IL, U.S.A.) was used for statistical analysis. T-test and Chi-squared test were used to determine statistical differences for ordinal and nominal measures. All statistical tests were performed using a 5% significance level (2-tailed). Regarding statistical analysis of the geographical distances, see Section 2.2.

3. Results

3.1. Epidemiology

The mean annual incidence of EOPD was 3.3/100,000 (range, 2.1–4.5/100,000) in the years 1995–2006, while that of PD was 25.2/100,000 (range, 22.2–28.2/100,000). The incidence of EOPD increased 1.7-fold between the years 1995–2006 (Fig. 1), while that of PD increased only 1.1-fold. Interestingly, there was a 1.3-fold increase in the incidence of PD in men, whereas no change was observed in the incidence of PD in women.

3.2. Clinical features of EOPD patients

The range of the age at onset (AAO) was 20–54 years and that of disease duration was 1–27 years among the 441 patients with EOPD. Delay from symptom onset to the PD diagnosis was 1.9 ± 2.2 years (mean $\pm$ standard deviation) and delay from the onset to initiation of medication was 2.1 ± 2.4 years. Three quarters of the patients were on L-dopa medication at the time of the study and

this drug had been in use for a mean time of 5.8 years. L-dopa treatment had been started 4.2 years after disease onset for the patients with AAO <50 years and 2.0 years after disease onset for the patients with AAO $\geq$ 50 years ($p < 0.001$). Male predominance was significant among patients with AAO <50 years (Table 1).

3.3. Clinical features of EOPD patients with family history

Family history of PD was reported by 30.6% of the patients (Table 2). Fifty-two patients (11.8%) reported an affected first-degree relative. The affected relative was the mother in 24 cases, the father in 17 cases, a sibling in 13 cases and a child in one case. Three patients had two affected first-degree relatives with PD including one patient with two affected siblings and two patients with an affected mother and a sibling. Segregation ratios were calculated to be 0.0465 for the parents and 0.0094 for the siblings. In total, we identified 1483 siblings and 29.5% of them were under 50 years of age. Siblings of patients with AAO $\geq$ 50 years were affected 2.7-fold more often than siblings of patients with AAO <50 years.

The 52 patients with an affected first-degree relative were compared with those who did not report any family history of PD ($N = 306$). The groups did not differ with respect to gender ratio, age at onset, age at medication or L-dopa initiation. Disease duration was 1.0 year longer and L-dopa was more frequently used among patients with affected first-degree relative (Table 1).

3.4. Geographical distribution of EOPD

Birthplaces were determined for 433 EOPD patients after the exclusion of five patients born abroad and three patients with an unreported place of birth. The birthplaces of patients with PD in first-degree relatives were concentrated in the western part of Finland and to a lesser extent in the province of Savo (Fig. 2A).

Table 2
Family history of PD among patients with EOPD.

	Total	AAO <50 y	AAO 50–54 y	Familial PD ^a	Sporadic PD
Patients, N	441	273	162	52	306
Family history of PD, % (N)					
Any relative affected	30.6 (135)	29.7 (81)	32.1 (52)	100 (52)	N.A.
First-degree relative affected	11.8 (52)	10.3 (28)	14.8 (24)	100 (52)	N.A.
Mother affected	5.4 (24)	5.1 (14)	6.2 (10)	46.2 (24)	N.A.
Father affected	3.9 (17)	3.4 (9)	5.1 (8)	32.7 (14)	N.A.
Sibling affected	2.9 (13)	1.8 (5)	4.9 (8)	25.0 (13)	N.A.
Parent habitual residence, % (N)					
Parents born in same municipality	35.1 (144)	34.0 (87)	36.9 (55)	46.2 (24)	33.8 (95)
Parents born in same province	58.2 (238)	57.0 (146)	60.5 (89)	65.4 (34)	58.4 (164)

^a Familial PD includes probands who had at least one first-degree relative affected by PD. Probands who had second or third-degree relative/-s affected by PD are not included in this table. Although exact AAO was not available for eight patients, two of them could be confidently classified to the age groups defined in the table. AAO, age at onset; N.A., not applicable.

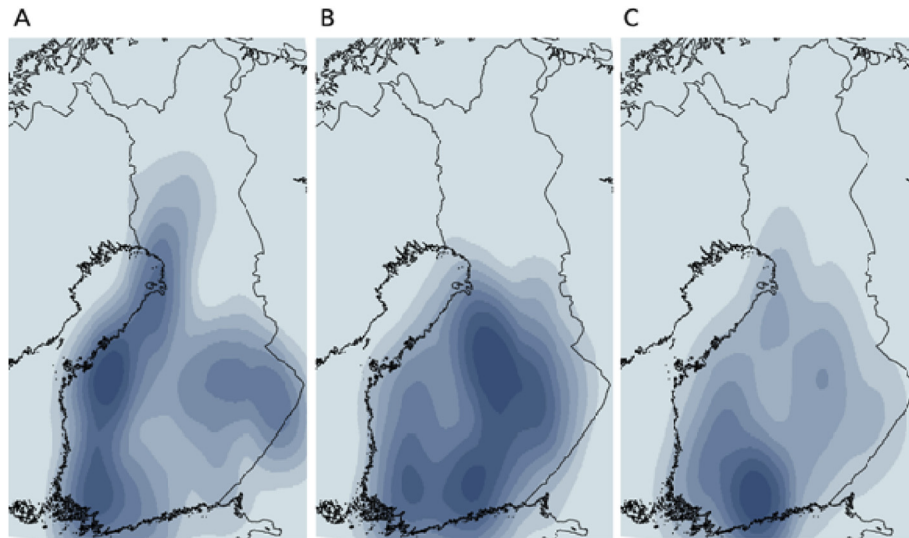


Fig. 2. Distribution of birthplaces of patients with PD, stratified according to their family history. A, patients with PD among first-degree relatives ($N = 52$). B, patients with PD among second-degree or more distant relatives only ($N = 80$). C, patients with no PD among first-degree or more distant relatives ($N = 301$).

Altogether, the birthplaces showed different clustering for patients with PD in first-degree relatives (Fig. 2A), patients with PD in second-degree or more distant relatives (Fig. 2B), and patients without known cases of PD among their relatives (Fig. 2C).

The distances between birthplaces of the parents could be determined for 49 patients with PD among first-degree relatives, 79 patients with PD among the more distant relatives, and 294 patients with negative family history. The distributions of distances were different between patients with negative family history and patients with PD among first-degree relatives ($p = 0.025$, K–S test) but not between patients with negative family history and patients with PD among their more distant relatives ($p = 0.98$, K–S test). The median distance was 9.3 km for patients with PD among first-degree relatives and 38 km for patients with negative family history ($p = 0.0077$ for difference, Wilcoxon rank-sum test). The median distance was also 38 km for patients with PD among the more distant relatives ($p = 0.48$ for difference in comparison with patients with negative family history, Wilcoxon rank-sum test). Q–Q plots indicated that short distances were overrepresented among the patients with PD among first-degree relatives as compared to the patients with negative family history (Supplemental fig. e–1A), but the same was not true for patients with PD among the more distant relatives (Supplemental fig. e–1B).

4. Discussion

We ascertained a nationwide cohort of EOPD patients by using a register of national health authorities. A total of 441 PD patients, with disease onset before 55 years, could be recruited and patient characteristics and family history were collected by means of a structured questionnaire. All the participants completed the questionnaires providing 98.0% of the information that was requested. Interestingly, the incidence of EOPD has increased between the years 1995–2006. First-degree relatives with PD were reported by 11.8% of the patients. The distribution of patients with a family history of PD was uneven across the country.

Our data suggested that the incidence of EOPD in Finland has increased since the year 1995. In addition, the mean crude incidence of PD was 25.2/100,000 between the years 1995 and 2006, being 1.5-fold higher than that in 1992 in Finland [3] and slightly higher than previous estimates of PD incidence in other populations [20]. Comparison of the incidence rates of PD in Finland between the

years 1971 and 1992 suggests a 1.2-fold increase [3]. A higher number of older people in the population is one factor contributing to the increase in incidence of PD, while unidentified environmental factors or more timely and more accurate diagnostics of patients with early-onset PD may contribute to the rise in incidence of EOPD. Many of the previous studies on incidence and prevalence of PD have been based on hospital registries or door-to-door surveys in a relatively small population. In this study we identified all PD patients who lived in Finland and who were granted reimbursement for PD medication. Patient identification was based on registry information and patient charts were not examined and, hence, occasional patients with misdiagnosed PD could have been included. However, patients reporting uncertainties in diagnosis in the questionnaire were excluded. On the other hand, towards the end of the study period we may have missed patients with incipient PD, who were not yet on PD medication.

Male predominance was seen in the entire study cohort and the difference in sex ratio was slightly more pronounced in patients with AAO below 50 years of age. Furthermore, the annual incidence of PD was higher in men between the years 1995 and 2011. Most epidemiological studies have reported a higher prevalence of PD in men [3,21–24].

Fifty-two patients (11.8%) reported an affected first-degree relative. This proportion is similar to those reported in previous studies on familial occurrence of PD [4–10]. Segregation ratios suggested autosomal dominant rather than recessive inheritance, but the ratios could be explained by incomplete penetrance or the fact that the siblings may not have reached the age of onset of PD. This finding differs from that in a previous study, where segregation ratios suggested recessive rather than dominant inheritance in the Finnish population [7]. This may be caused by the fact that age was not a selection criterion in the previous study, whereas in the present study only early-onset patients were recruited. The mean age of the patients was 70.5 years in the previous study [7], whereas that in the present study was 55.1 years.

Within Finland, a relatively strong autosomal substructure defining eastern and western populations has been suggested [25]. Interestingly, we found that the birthplaces of EOPD patients are distributed unevenly across the country. The birthplaces of patients with PD among first-degree relatives were clustered in south-western and western coastal provinces and in the eastern province of Savo suggesting that monogenic forms of PD or genetic PD

susceptibility are present in the population in these areas. Such a clustering would support autosomal dominant inheritance or a shared environmental factor. The birthplaces of patients without a family history were more or less clustered in southern Finland reflecting the higher population density there than elsewhere in the country. A genetic contribution was further supported by the finding that the distance between the birthplaces of the parents was shorter for patients with PD among their first-degree relatives than for patients with negative family history. The close proximity of the parental birthplaces intuitively correlates with increased inbreeding coefficient, and the presence of autosomal recessive susceptibility genes for PD in the population could provide an explanation for our findings. The clustering of birthplaces of the parents supports a strong genetic component in the etiology of familial EOPD. Since known genetic causes of PD were not excluded, mutations may partly explain the familial clustering found in the population.

The nationwide cohort enabled us to study the epidemiology of EOPD. We found that incidence of EOPD has increased during a 12-year period. An affected first-degree relative was reported by 11.8% of the probands and the segregation ratios suggested autosomal dominant rather than recessive inheritance. Interestingly, the birthplaces of EOPD patients were unevenly distributed across the country and the distance between the birthplaces of the parents was shorter for patients with PD among their first-degree relatives than for patients without a family history. These findings suggest that monogenic forms of PD or genetic PD susceptibility are present in the population, but such genes remain to be detected.

Author contributions

P. Ylikotila: data collection and analysis, statistical analysis, writing the manuscript. T. Tiirikka: data analysis, statistical analysis. J. Moilanen: data analysis. K. Majamaa: study design, data analysis, statistical analysis, writing the manuscript. All authors contributed to the conception of the study, revising the manuscript for intellectual content and approved the final version for publication.

Conflict of interest

None.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.parkreldis.2015.06.003>.

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